SENTINEL-X

Digital Forensics & Incident Investigation Report

Case: DEMO

Generated: 2026-11-10T12:22:27+00:00

Tool: SENTINEL-X 0.2.0

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1. Executive Summary

159 web requests from 5 unique IP addresses were analyzed (2026-11-03 08:00:10Z to 2026-11-03 10:58:49Z). 18 findings were produced: 4 Medium, 4 Low, 8 High, 2 Critical.

Initial access: Most likely initial access (hypothesis): Exploitation of a web application (SQL injection) from 198.51.100.23. Confidence: Medium Confidence. This is a hypothesis derived from the findings, not a confirmed fact.

Classification legend - Observed: directly present in evidence. Inferred: conclusion from several indicators. Unknown: cannot be established with the available evidence. Severity (impact) and confidence (strength of evidence) are independent.

18	10	1	2
FINDINGS	CRITICAL + HIGH	SOURCE IPs	EVIDENCE FILES

2. Evidence Inventory & SHA-256

ID	File	Format	Size	SHA-256	Collected (UTC)
EV-0001	access.log	web	21276	59f4fccc8d305d424cbea3a69c9a868d6ca86d1affdae66359e3f26d3a57ed10	2026-11-10 12:22:27Z
EV-0002	auth.log	auth	2130	7bd4c3d12d8b7f1aeca292dcfe2f61668d8216e550dfc9bf85fa56ca7dfee3bd	2026-11-10 12:22:27Z

Integrity check before analysis: all SHA-256 hashes matched.

3. Initial Access Analysis

Most likely initial access (hypothesis): Exploitation of a web application (SQL injection) from 198.51.100.23

First suspicious event: F-0008 at 2026-11-03 10:10:00Z from 198.51.100.23 targeting /products.php?id=3'+OR+1=1-- (EV-0001, line 113).

Supporting findings: F-0008, F-0009, F-0010, F-0012, F-0013, F-0014, F-0018

Contradicting / limiting evidence: none identified

Missing evidence: Application and database logs (to confirm exploit success); File integrity data or reference hashes (new/modified files); Process, cron and network-flow records (post-exploitation activity)

Confirmed (Observed) login after failures: F-0018 user 'deploy' from 198.51.100.23 at 2026-11-03 10:30:40Z (EV-0002:19).

Attack path reconstruction

Stages are ordered by first-seen time; a chain is a reconstruction aid, not proof that one actor did every step.

4. Attack Timeline

Time	Phase	Finding	IP	Summary
2026-11-03 10:00:00Z	Reconnaissance	F-0001 (EV-0001:81)	198.51.100.23	22 HTTP 404 responses from one IP; >= 15 within 60s (directory/content discovery pattern).
2026-11-03 10:00:00Z	Reconnaissance	F-0002 (EV-0001:81)	198.51.100.23	User-Agent matches known scanning/attack tool 'nikto' (first occurrence).
2026-11-03 10:00:01Z	Reconnaissance	F-0003 (EV-0001:82)	198.51.100.23	Request targets a sensitive path (config, VCS, backup, admin tooling).
2026-11-03 10:00:02Z	Reconnaissance	F-0004 (EV-0001:84)	198.51.100.23	Request targets a sensitive path (config, VCS, backup, admin tooling).

Time	Phase	Finding	IP	Summary
2026-11-03 10:00:03Z	Reconnaissance	F-0005 (EV-0001:85)	198.51.100.23	Request targets a sensitive path (config, VCS, backup, admin tooling).
2026-11-03 10:00:04Z	Reconnaissance	F-0006 (EV-0001:86)	198.51.100.23	Request targets a sensitive path (config, VCS, backup, admin tooling).
2026-11-03 10:10:00Z	Reconnaissance	F-0007 (EV-0001:113)	198.51.100.23	User-Agent matches known scanning/attack tool 'sqlmap' (first occurrence).
2026-11-03 10:10:00Z	Initial Access / Exploitation	F-0008 (EV-0001:113)	198.51.100.23	Request contains SQL injection patterns.
2026-11-03 10:10:20Z	Initial Access / Exploitation	F-0009 (EV-0001:114)	198.51.100.23	Request contains SQL injection patterns.
2026-11-03 10:10:40Z	Initial Access / Exploitation	F-0010 (EV-0001:115)	198.51.100.23	Request contains path traversal patterns.
2026-11-03 10:15:00Z	Authentication	F-0011 (EV-0001:119)	198.51.100.23	8 failed login responses (401/403) from one IP; >= 5 within 300s.
2026-11-03 10:16:10Z	Authentication	F-0012 (EV-0001:128)	198.51.100.23	Login-path POST returned 302 after 8 failures from the same IP.
2026-11-03 10:19:10Z	Execution / Persistence indicator	F-0013 (EV-0001:131)	198.51.100.23	Script request with command-execution parameters or web-shell-like name.
2026-11-03 10:19:40Z	Execution / Persistence indicator	F-0014 (EV-0001:132)	198.51.100.23	Script request with command-execution parameters or web-shell-like name.
2026-11-03 10:30:00Z	Reconnaissance	F-0015 (EV-0002:5)	198.51.100.23	4 distinct non-existent usernames tried from one IP (admin, oracle, test, ubuntu).
2026-11-03 10:30:00Z	Authentication	F-0016 (EV-0002:6)	198.51.100.23	10 failed SSH authentications from one IP; >= 5 within 300s.
2026-11-03 10:30:40Z	Confirmed Access	F-0018 (EV-0002:19)	198.51.100.23	sshd logged a SUCCESSFUL login for 'deploy' (password) after 10 failed attempts from the same IP.

5. Findings & Confidence Levels

ID	Type	Time	Evidence	IP	Sev.	Confidence / Class	Reason
F-0001	content_enumeration	2026-11-03 10:00:00Z	EV-0001:81	198.51.100.23	Medium	High Confidence / Observed	22 HTTP 404 responses from one IP; >= 15 within 60s (directory/content discovery pattern).
F-0002	scanner_user_agent	2026-11-03 10:00:00Z	EV-0001:81	198.51.100.23	Medium	High Confidence / Observed	User-Agent matches known scanning/attack tool 'nikto' (first occurrence).
F-0003	sensitive_path_access	2026-11-03 10:00:01Z	EV-0001:82	198.51.100.23	Low	Medium Confidence / Observed	Request targets a sensitive path (config, VCS, backup, admin tooling). Response status 404 suggests it did not succeed (not proven).
F-0004	sensitive_path_access	2026-11-03 10:00:02Z	EV-0001:84	198.51.100.23	Low	Medium Confidence / Observed	Request targets a sensitive path (config, VCS, backup, admin tooling). Response status 403 suggests it did not succeed (not proven).
F-0005	sensitive_path_access	2026-11-03 10:00:03Z	EV-0001:85	198.51.100.23	Low	Medium Confidence / Observed	Request targets a sensitive path (config, VCS, backup, admin tooling). Response status 403 suggests it did not succeed (not proven).
F-0006	sensitive_path_access	2026-11-03 10:00:04Z	EV-0001:86	198.51.100.23	Low	Medium Confidence / Observed	Request targets a sensitive path (config, VCS, backup, admin tooling). Response status 403 suggests it did not succeed (not proven).
F-0007	scanner_user_agent	2026-11-03 10:10:00Z	EV-0001:113	198.51.100.23	Medium	High Confidence / Observed	User-Agent matches known scanning/attack tool 'sqlmap' (first occurrence).
F-0008	sql_injection_attempt	2026-11-03 10:10:00Z	EV-0001:113	198.51.100.23	High	Medium Confidence / Observed	Request contains SQL injection patterns. Response status 500 suggests it did not succeed (not proven).
F-0009	sql_injection_attempt	2026-11-03 10:10:20Z	EV-0001:114	198.51.100.23	High	Medium Confidence / Observed	Request contains SQL injection patterns. Response status 200 alone does not prove the attempt succeeded.
F-0010	path_traversal_attempt	2026-11-03 10:10:40Z	EV-0001:115	198.51.100.23	High	High Confidence / Observed	Request contains path traversal patterns. Response status 200 alone does not prove the attempt succeeded.
F-0011	bruteforce_login_attempts	2026-11-03 10:15:00Z	EV-0001:119	198.51.100.23	High	High Confidence / Observed	8 failed login responses (401/403) from one IP; >= 5 within 300s.
F-0012	possible_login_success_after_failures	2026-11-03 10:16:10Z	EV-0001:128	198.51.100.23	High	Medium Confidence / Inferred	Login-path POST returned 302 after 8 failures from the same IP. Web logs cannot confirm authentication; verify in application/auth logs.
F-0013	possible_web_shell	2026-11-03 10:19:10Z	EV-0001:131	198.51.100.23	Critical	Medium Confidence / Inferred	Script request with command-execution parameters or web-shell-like name. Response status 200 alone does not prove the attempt succeeded.
F-0014	possible_web_shell	2026-11-03 10:19:40Z	EV-0001:132	198.51.100.23	Critical	Medium Confidence / Inferred	Script request with command-execution parameters or web-shell-like name. Response status 200 alone does not prove the attempt succeeded.

ID	Type	Time	Evidence	IP	Sev.	Confidence / Class	Reason
F-0015	ssh_user_enumeration	2026-11-03 10:30:00Z	EV-0002:5	198.51.100.23	Medium	Medium Confidence / Observed	4 distinct non-existent usernames tried from one IP (admin, oracle, test, ubuntu).
F-0016	ssh_bruteforce_attempts	2026-11-03 10:30:00Z	EV-0002:6	198.51.100.23	High	High Confidence / Observed	10 failed SSH authentications from one IP; >= 5 within 300s. Accounts tried: admin, deploy, oracle, root, test, ubuntu.
F-0017	cross_source_correlation	2026-11-03 10:30:40Z	EV-0002:19	198.51.100.23	High	Medium Confidence / Inferred	IP 198.51.100.23 appears in web attack activity (first: EV-0001:81, scanner_user_agent) and also has a logged successful SSH login (EV-0002:19). The same actor is plausible but not proven (shared or NAT addresses are possible).
F-0018	ssh_login_success_after_failures	2026-11-03 10:30:40Z	EV-0002:19	198.51.100.23	High	High Confidence / Observed	sshd logged a SUCCESSFUL login for 'deploy' (password) after 10 failed attempts from the same IP. The login is recorded fact; attribution of the actor and intent remain inferred.

6. Indicators of Compromise (local extraction)

IP addresses

Value	Related findings
198.51.100.23	F-0001, F-0002, F-0003, F-0004, F-0005, F-0006, F-0007, F-0008, F-0009, F-0010, F-0011, F-0012, F-0013, F-0014, F-0015, F-0016, F-0017, F-0018

User-Agents

Value	Related findings
Mozilla/5.0 (X11; Linux x86_64) Firefox/126.0	F-0011, F-0012, F-0013, F-0014
Mozilla/5.00 (Nikto/2.5.0) (Evasions:None) (Test:Port Scanner)	F-0001, F-0002, F-0003, F-0004, F-0005, F-0006
sqlmap/1.8	F-0007, F-0008, F-0009, F-0010

URLs / paths

Value	Related findings
/.env	F-0004
/.git/config	F-0005
/admin	F-0001, F-0002
/backup	F-0003
/download.php	F-0010
/phpmyadmin/	F-0006
/products.php	F-0007, F-0008, F-0009
/wp-content/uploads/2026/11/shell.php	F-0013, F-0014
/wp-login.php	F-0011, F-0012

An IP address does not identify a person; geolocation/ASN enrichment is not part of the MVP.

7. Remediation Recommendations

Priority	Action	Findings
Immediate	Preserve evidence first: snapshot disk/memory and keep the original logs before any cleanup.	

Priority	Action	Findings
Immediate	Disable script execution in upload directories; compare web root against a clean baseline.	F-0013, F-0014
Immediate	Force a password reset for the affected account(s) and review activity after the login time.	F-0012
Immediate	Isolate the host from the network if a web shell is suspected; locate and hash the file, do not execute it.	F-0013, F-0014
Immediate	Terminate the session, reset the credentials and review ~/.ssh/authorized_keys, cron jobs, new users and services.	F-0018
Immediate	Treat as a potential compromise: extend the investigation to other hosts and block the IP after preserving evidence (the IP may be shared).	F-0017
High	Disable SSH password authentication (key-only), add fail2ban or equivalent rate limiting.	F-0016
High	Enable rate limiting / lockout and MFA on the login endpoint.	F-0011
High	Normalize and validate file parameters; restrict the web user's filesystem access (least privilege).	F-0010
High	Review commands and file changes made by that account after the login time (shell history, auditd).	F-0018
High	Review database logs for the same timeframe to see whether queries executed.	F-0008, F-0009
High	Review the targeted endpoint; use parameterized queries and input validation; add a WAF rule.	F-0008, F-0009
High	Rotate secrets reachable by the web server user (DB passwords, API keys).	F-0013, F-0014
Medium	Block known scanner User-Agents at the edge and monitor the same IP for follow-up activity.	F-0002, F-0007
Medium	Rate-limit or block the source after preserving evidence; verify no sensitive files are exposed.	F-0001
Medium	Remove backups, VCS folders and config files from the web root; restrict admin tools by IP.	F-0003, F-0004, F-0005, F-0006
Medium	Remove unused accounts and restrict allowed users with AllowUsers/AllowGroups.	F-0015
Medium	Restrict SSH exposure by source IP / VPN.	F-0016

Generic defensive guidance derived from finding types; validate in your environment.

8. AI Investigation Summary (optional, unverified unless checked)

Not generated (AI step not run or Ollama unavailable). All other sections are rule-based.

9. Limitations

- Web access logs show requests and response codes only; they cannot prove that an exploit or login succeeded.

- An IP address does not identify a person. Geolocation and VPN/proxy status are not assessed in this version.

- Findings are rule-based and may include false positives or miss unknown techniques.

- Only Apache/Nginx access logs and SSH auth logs are analyzed in this version; other evidence types are listed as unsupported.

- Classic syslog timestamps have no year or timezone: the year comes from --year or the file modification time and times are treated as UTC.